

Reel or Real? A Context-Aware Deepfake Detection and Verification Framework

1. Introduction

Deepfakes – highly realistic but fake videos or audio generated by AI – have rapidly evolved from a novelty to a serious threat to information integrity. Recent studies highlight a dramatic rise in the realism of deepfakes and the corresponding difficulty of detecting them, even with state-of-the-art methods. This escalation poses risks across society: malicious deepfakes have been implicated in misinformation during elections and in harassment and identity fraud cases. Public concern is high – over 90% of adults in one survey feared deepfake misinformation – yet fewer than 10% could confidently spot a fake. This gap between the threat and the public's ability to discern truth underscores the urgent need for reliable detection tools.

Academia and industry have responded with a surge of deepfake detection research, but significant limitations remain. Traditional detectors often perform well only on the specific data they were trained on and fail to generalize to new videos or manipulation techniques. Moreover, many AI models act as “black boxes,” giving a verdict without explanation. Such opaque systems struggle to earn user trust or meet legal/ethical scrutiny. Consequently, recent scholarship emphasizes that detection systems must not only be accurate, but also generalizable, multimodal, and explainable. In other words, a real-world solution should handle a diverse range of fakes, integrate multiple evidence sources (visual, audio, metadata), and clearly communicate why a piece of content is flagged as fake.

This project aims to build a context-aware deepfake detection and verification framework that meets these needs. Context-aware here means the system goes beyond analyzing pixels: it will fuse visual and audio cues, inspect metadata for authenticity, and consider the source context of the media. Uniquely, the project also incorporates an educational component – providing a sandbox for users to experiment with creating and detecting deepfakes – to raise awareness and literacy. By combining robust AI detection with user-centered design, transparency, and education, the system is intended for deployment in real-world settings where both accuracy and understanding are paramount. The following sections outline the relevant research context and the methodology behind developing this system.

2. Literature Review

Social Impact and the Need for Trustworthy Detection

The proliferation of deepfakes has far-reaching social and ethical implications. Studies across continents report that while people are aware of deepfakes and express high anxiety about them, they generally struggle

to identify manipulated media in practice. This vulnerability has been exploited in various harmful ways – from false news videos affecting democratic discourse to AI-generated pornography used for harassment. The erosion of trust in digital content is a key concern: if any video could be fake, public confidence in legitimate information can crumble. Thus, researchers argue that detection technology must be trustworthy and accessible to non-experts. A simple true/false label is not enough; the system should explain its reasoning in an intuitive way. Sippy et al. (2024) found that users are far more likely to trust a deepfake detector's verdict if they can see visual evidence or rationale for that decision. In response, recent approaches pair detection with user-facing explanations and provenance indicators to help rebuild digital trust. Crucially, scholars frame deepfake detection not just as a technical arms race, but as an ethical and educational mission – one that requires improving public media literacy and involving stakeholders like journalists, educators, and policymakers in the loop. These insights reinforce our project's dual emphasis on detection performance and user education.

Evolution of Deepfake Detection Techniques

Early deepfake detectors were based on Convolutional Neural Networks (CNNs) trained to spot visual anomalies. Classic methods looked for tell-tale signs of tampering – e.g. unnatural facial blurring, color mismatches, or inconsistent blinking patterns. In controlled tests, CNN models like Xception, VGG, or ResNet achieved impressively high accuracy (up to ~95%) on curated deepfake datasets. However, these models often overfit to the specific artifacts of the training set and “degraded sharply” when confronted with new types of forgeries or real-world videos. This brittleness became evident as deepfakes grew more sophisticated and diverse. Researchers responded by exploring more generalized architectures: transformer-based models and hybrid CNN–transformer networks that capture temporal dynamics and contextual cues beyond single frames. For example, Gong and Li (2024) categorize modern detectors into CNN, transformer, and even biological signal-based classes, noting that hybrids tend to be more robust across varied data. One innovative approach, MFCLIP (Zhang et al., 2025), extends a vision-language transformer to detect AI-generated imagery, enabling identification of deepfakes created by the latest text-to-image diffusion models. Another emerging direction uses physiological signals: subtle human micro-expressions or heart rate patterns that are hard for deepfake generators to faithfully reproduce. Techniques tracking eye-blink irregularities or pulse signals in videos have shown promise in spotting imperceptible fake cues. Overall, the field's trajectory is clear – moving toward multimodal, context-aware detectors that combine visual, audio, and even biometric hints to improve generalization. This project's design builds on that trajectory, employing a hybrid model that leverages both CNN and transformer components for spatial and temporal features, along with any auxiliary cues that can strengthen detection.

Dataset Diversity and Generalization Challenges

A consistent lesson from the literature is that “robust detection depends fundamentally on dataset diversity.” Deepfake models can fail catastrophically outside their training domain, so using a wide variety of training data is crucial. Early benchmark datasets, such as FaceForensics++ or Facebook’s Deepfake Detection Challenge (DFDC), broke new ground for the community but had limitations: they often contained easily detectable fakes or lacked demographic diversity. Models trained on those homogeneous datasets tended to perform well in lab settings yet stumbled on real-world media. In response, newer datasets greatly expanded realism and scope. The Korean Deepfake Dataset (KoDF) introduced 175,000 deepfake videos of Korean actors to counter the Western-centric bias in prior data. WildDeepfake, meanwhile, collected actual illicit deepfake videos “in the wild,” complete with the kind of compression and noise found on social platforms. Evaluations showed that if a detector was trained on one dataset and tested on a completely different one, its accuracy could drop by over 20 percentage points – a dramatic generalization gap. This has driven the community to keep pushing for broader and more inclusive corpora. Multilingual and multimodal datasets are also emerging. For instance, FakeAVCeleb combined manipulated video and cloned audio for the same subjects, demonstrating that an audio-visual detector can outperform a video-only model by a significant margin. (Khalid et al. (2021) report a boost in F1-score to ~94% with fused audio-video features, more than 10% higher than visual-only.) Likewise, researchers have created deepfake datasets for different languages and contexts (e.g., a recent Hindi deepfake dataset HAV-DF in 2024) to ensure detection methods generalize across linguistic and cultural boundaries. The clear consensus is that a detection model must be trained on diverse, representative, and realistic data to be dependable in the real world. This insight directly informs our project’s data strategy – we prioritize variety in our training data and plan rigorous cross-dataset testing to guard against brittle performance.

Explainability, Provenance, and User Trust

No matter how accurate an AI model is, it will not be fully effective if users and decision-makers do not trust it. Thus, explainability and media provenance have become central research themes in deepfake detection. Lyu (2020) warned early that purely black-box neural networks, which simply output “real” or “fake,” “lack the interpretability required for journalism, law, or policy enforcement”. In sensitive use-cases (e.g. verifying a news video or a legal piece of evidence), stakeholders need to understand why a clip was flagged. To address this, many researchers integrate explainable AI techniques: for example, attention-weighted heatmaps or Grad-CAM visualizations can highlight which regions of a face were deemed manipulated. Such visual cues help users see the same inconsistencies the model sees (for instance, unnatural eye textures or warped teeth in a fake video). User studies indicate this transparency directly impacts trust – when users “can see why content is flagged, they

are more likely to believe the system”. Beyond AI explainability, the notion of provenance – tracking the origin and authenticity of media – has gained traction. Rather than only reacting to fakes, a complementary approach is to proactively certify real media. One emerging standard is the C2PA (Coalition for Content Provenance and Authenticity), which allows publishers to embed tamper-evident metadata and cryptographic signatures into content at creation. If a video has a valid provenance trail (originating from a trusted camera with no edits), a detector can use that as strong evidence of authenticity; conversely, missing or mismatched metadata could raise suspicion. Researchers have even proposed blockchain-based registries and digital watermarks for videos and audio as additional authenticity checks. Our project sits at the intersection of these trends: it not only uses explainable AI to show how it detects fakes but also incorporates metadata analysis to verify where a video comes from. By combining detection with provenance verification, the framework aims to shift from a purely “fake-spotting” tool to a more comprehensive truth-verification system. This aligns with the latest thinking that deepfake prevention requires both robust technical detection and built-in measures for content authenticity.

3. Proposed System Design and Methodology

To address the challenges identified above, we have designed a modular system with multiple integrated components. The emphasis is on replicating real-world conditions as closely as possible in both the data and the detection approach, while also fostering user understanding. Key aspects of our methodology include the dataset strategy and several novel feature modules, described below.

Dataset Strategy and Replacement

Realistic and diverse training data is the backbone of our approach. Initially, our plan was to utilize the Facebook Deepfake Detection Challenge (DFDC) dataset and the KoDF dataset as primary sources, due to their large scale and diversity (DFDC has thousands of videos of various actors; KoDF offers extensive non-Western content). However, due to access and licensing issues, we could not obtain DFDC and KoDF as originally intended. In their place, we conducted a careful selection of alternative datasets that collectively match or exceed the scope of the originals in size, diversity, and realism. The revised dataset lineup is:

- Celeb-DF-v2: A large-scale public deepfake video dataset featuring celebrity faces, known for high visual realism and challenging fake artifacts. This dataset plays a similar role to DFDC in providing a wide variety of subjects and scenarios, ensuring the model sees many different identities and fake generation techniques.
- DeeperForensics-1.0: A rigorously engineered benchmark from 2020 that contains high-quality manipulated videos with diverse perturbations (like various lighting, compression levels, and occlusions). DeeperForensics

complements Celeb-DF-v2 by introducing controlled degradations, helping the system become resilient to real-world video distortions.

- **FaceForensics++:** One of the early standard datasets, included here to ensure coverage of traditional fake generation methods (e.g., face swaps and expression transfers) on relatively controlled footage. FaceForensics++ provides a baseline for classic fake patterns and also allows comparison of our results with prior work, since it's a common benchmark.
- **FakeAVCeleb:** An audio-visual deepfake dataset where not only the video is fake, but the voices are cloned as well. We retained FakeAVCeleb from the original plan because it specifically tests multimodal detection – models trained on it have to verify if the speaker's face and the voice are from the same person. Including this dataset aligns with research that showed audio-visual fusion improves detection significantly and addresses the underdeveloped area of audio consistency checking in deepfake detection.
- **Real-World Collected Set:** In addition to public datasets, we created a bespoke dataset via a university-approved survey and crowdsourcing effort. We gathered videos and images from 369 volunteer participants, representing "in-the-wild" content. This set includes genuine content and some participant-generated deepfakes. Its purpose is to inject real-world variability (different devices, resolutions, social media artifacts) and to evaluate the framework on completely genuine vs. fake examples reflective of what everyday users might encounter. All necessary consent and ethical approvals were obtained for this data collection, and it serves as an invaluable testbed for generalization beyond academic datasets.

By combining these five sources, we ensure the training data spans multiple languages, demographics, and fake-generation methods. The scale is on par with (or larger than) DFDC+KoDF combined, preserving the original goal of a broad training corpus. We anticipate that models trained on this composite dataset will maintain strong cross-dataset generalization – a hypothesis we will validate by holding out certain datasets for testing only. In essence, although the exact datasets changed, the project's commitment to data diversity and realism remains firm: this strategy is directly aimed at preventing the overfitting and brittleness that prior detectors suffered when confronted with new input.

System Architecture and Key Modules

The system's architecture follows a modular hybrid design, comprising a deep learning core and several supporting components. At its heart is a two-branch neural network: one branch processes video frames (spatial analysis) and the other processes audio (sound analysis). A temporal transformer module then fuses information across frames (and audio timestamps) to detect inconsistencies over time. Surrounding this core are modules that analyze contextual data and enhance the interpretability of results. Below we detail the major features/modules already planned or under development:

- **Metadata Provenance Analysis:** Beyond the pixels and waveforms, the system examines the metadata associated with a video or image. This includes standard EXIF metadata (camera model, timestamps, software used) as well as any available platform data (e.g. social media timestamps, upload source, digital signatures). Anomalies or inconsistencies in metadata can tip off forgeries – for example, if a video claiming to be “unedited” shows signs of having been processed by editing software in its metadata. Our tool will automatically flag such clues. This approach is inspired by emerging content authentication standards like C2PA, which embed tamper-evident metadata and cryptographic signatures into media files. By comparing a file’s metadata against expected values or known trusted signatures, the system adds a layer of provenance verification. In practice, if a deepfake video lacks a legitimate metadata trail (or if its hashes don’t match any registered authentic asset), the system will treat it with higher suspicion. Metadata analysis thus provides a form of “digital chain-of-custody” checking, complementing the AI’s judgment with a simpler, rule-based authenticity test.
- **Multimodal Audio-Video Fusion:** The core detection model uses a fusion of audio and visual signals to identify deepfakes. This module ensures that the speaking face in the video and the accompanying voice are coherent and genuinely paired. It extracts audio features (like MFCC – Mel-Frequency Cepstral Coefficients – representing the vocal characteristics) and visual features (from the CNN processing frames) and then aligns them over time. Any misalignment – e.g., lips moving out of sync with spoken words, or an obvious mismatch between the face and voice identity – can indicate a deepfake. Incorporating multimodal analysis is motivated by research findings that audio-visual cues together significantly boost detection accuracy. Many deepfake videos only manipulate the face and not the voice (or vice versa), so a unimodal detector might miss those; a fused approach catches discrepancies between modalities. Our design addresses a noted gap in prior systems, which often ignored audio entirely. By doing so, we expect to improve generalization (since even if a fake perfectly mimics the visual style, it might still slip up on the audio, or vice versa).
- **Facial Micro-Expression Analysis:** Deepfake algorithms, while incredibly advanced, still struggle to recreate the full range of subtle human expressions and idiosyncratic movements. This module focuses on detecting those “tells.” It monitors fine-grained facial dynamics such as eye blinking frequency and synchronization, the natural motion of eyebrows, slight head tilts, lip and jaw movements when speaking, and even eye gaze stability. Human faces have subconscious micro-expressions and rhythms – for instance, real people blink at irregular intervals and have unique eyebrow raise patterns. Deepfakes often produce more mechanical or unnatural rhythms (early deepfakes infamously had almost no blinking at all, and even modern ones can have minor glitches in expression timing). By quantitatively modeling these behaviors, our system adds another discriminative signal. Prior work on biological and physiological cues supports this approach:

Heidari et al. (2023) showed that monitoring signals like heart rate or subtle eye-muscle movements can reveal imperceptible discrepancies in synthetic videos. Our implementation will use temporal analysis techniques (possibly optical flow or keypoint trajectory analysis) to spot when a subject's micro-expressions don't follow expected human patterns. This is especially useful for detecting "low-quality" deepfakes that may not have obvious visual artifacts but still exhibit odd timing or stiffness in facial motions.

- **Educational Deepfake Creation Sandbox:** In line with the project's educational mission, we are developing a sandbox environment where users can try creating their own deepfakes and then run our detector on them. The rationale is that hands-on experience can demystify how deepfakes work, and by seeing the process (and limitations) of generation, users learn what telltale signs to look for. The sandbox will be a web-based interface (part of the frontend) allowing a user to upload two face images or a short video and audio clip, and then apply an off-the-shelf face-swapping algorithm to produce a fake. It will then show the detection result and explanation for this generated fake. This feature is meant for educational and demonstration purposes: for instance, journalists or students can safely experiment with deepfake generation without needing technical expertise. By observing how even simple deepfakes can fool the eye but still be caught by certain analyses, users gain insight into why detection is non-trivial. Importantly, this interactive module addresses the call in literature for greater AI literacy and public engagement regarding deepfakes. Rather than just presenting a tool that labels content, we provide a sandbox to teach why content might be fake. All sandbox operations will happen locally in the browser or on the server without publishing any created fake, to avoid misuse. It is strictly an educational toy, reinforcing the system's philosophy that empowering users is as important as catching fakes.
- **Explainable AI Feedback (Grad-CAM & Saliency Reports):** To ensure transparency, every detection decision made by our AI will be accompanied by an explanation module output. This feature fulfills the project's core principle that a "user-centred design" is required for a deepfake detector. Concretely, for any video analyzed, the system will display visual heatmaps over one or more video frames, using techniques like Grad-CAM (Gradient-weighted Class Activation Mapping) to highlight which regions of the image the model found most indicative of a deepfake. For example, if the detector focuses on the boundary of the face and neck, the heatmap will illuminate that area, suggesting a blending or splicing artifact was key to the decision. Likewise, if an unusual eye region texture raised the model's suspicion, the user will see the eyes highlighted on the output frame. Alongside heatmaps, we provide a short text or graphical summary of other salient signals – such as an audio confidence score (did the voice match the face?) or a plot of blinking frequency over time compared to normal ranges. By presenting this information, the system effectively "communicates why a video is fake rather than simply labeling it as such." This approach directly

responds to the literature's caution that even a highly accurate tool may fail to inspire confidence if it cannot explain itself. We expect these explainability features to increase user trust and understanding, as supported by prior user studies. In professional settings (e.g., a fact-checker or law enforcement using our tool), the explainability also adds legal and ethical robustness – the system is not a mysterious oracle, but a transparent framework providing evidence for its conclusions. We will leverage techniques from recent research to refine these explanations, possibly including transformer attention visualizations (to show temporal reasoning) or layering the metadata verification results (so a user can see, for instance, “this video was flagged fake and it has no valid source signature”). Ultimately, the explainability module is what turns our detection system into a usable verification tool, bridging the gap between AI decision-making and human judgment.

All these modules work in concert as a pipeline. A user uploads a video through the web interface; the backend will run metadata checks, split the video into frames and audio, run the deepfake detection model on those inputs, perform the multimodal consistency analysis and micro-expression checks, and then output both a verdict (real or fake with a confidence score) and an explanation report. The user can then explore the evidence via the interface and even experiment in the sandbox mode to further learn about deepfakes. Each design choice – from integrating audio analysis to providing a sandbox – has been made to enhance either the accuracy of the system or the usability/trust for the user, or both. Notably, these decisions have been vetted through consultations with top AI professionals and domain experts. Discussions with experts in machine learning and digital forensics helped validate that our chosen approach addresses real-world needs. For example, industry professionals emphasised the importance of metadata provenance (a feature sometimes overlooked in purely academic projects) and helped refine our plan for the C2PA-based checks. Likewise, feedback from an AI ethics researcher reinforced our commitment to explainability and influenced how we present the results to users. This collaborative, well-researched design process gives us confidence that our system will not only be technically sound but also valuable.

4. Current Progress and Roadmap

As of early January 2026, the project is progressing well and remains on schedule according to the plan. During the initial phase (Oct–Dec 2025), we focused on research, design, and proof-of-concept developments. We completed an extensive literature review and problem scoping exercise by mid-November, which informed the system design and data choices described above. By late November, we had defined clear functional requirements and a detailed system architecture, including module specifications and API designs for the backend and frontend. In early December, we implemented a small baseline model test: a prototype CNN was trained on a subset of FaceForensics++ to ensure our data pipeline and training code were functioning correctly. This baseline experiment, as

expected, achieved high accuracy on easy fakes but highlighted the generalization issues noted in literature, reinforcing the need for our multimodal, multi-dataset approach (an important sanity check for the project).

Throughout December, work shifted to authoring the Contextual Report. We drafted the methodology and design sections (a refined version of the project proposal with added depth from new literature) by mid-December, and a polished full draft of the report – including introduction, literature review, and methodology – was completed by the end of December. This timing aligns with Week 13 of our plan, allowing for final proofreading and supervisor feedback in the first half of January. We are on track to submit the finalized contextual report by the project deadline (mid-January 2026), satisfying the Phase 2 goals.

On the development side, we have already begun building the actual system components in parallel with the research work. A preliminary data processing pipeline is in place: we have scripts for extracting video frames, normalizing image data, and computing audio features (e.g., MFCCs) from input videos. These have been evaluated on samples from our chosen datasets to ensure compatibility and correctness. The backend of the system is being developed using FastAPI (Python), and a basic REST API now exists locally to accept a video upload and return a dummy analysis result. This backend will soon integrate our trained models and modules as they become available. Meanwhile, a prototype frontend has been created using Next.js (React/TypeScript). The web interface currently supports uploading a video, showing a placeholder for results, and even includes a stub of the educational sandbox interface (for example, we have a UI form ready for the face-swapping demo, which will be connected to the backend generation code later). We have run the integrated frontend-backend locally, confirming that the pieces communicate (e.g., videos uploaded via the web UI are received by the FastAPI server correctly). In short, the groundwork for the full application is laid, and we are now transitioning into the core implementation and testing phase (Phase 3 of the project plan).

In the coming weeks (Jan-Mar 2026), our focus will be on model training, system integration, and evaluation. According to the roadmap, the next immediate step is scaling up the data pipeline to handle the entire datasets and training the first version of our visual deepfake detection model (using, say, an Xception or EfficientNet backbone on frame data) followed by the audio model and then the fused multimodal model. We have access to a GPU workstation for training, and we may leverage cloud resources for heavier training runs if needed. As models reach acceptable performance on validation sets, we will integrate them into the FastAPI backend, replacing the dummy components. By mid-project, we expect to deploy the system on a cloud server (or university server) for broader testing – this will allow us to simulate real user access, evaluate the system's load handling, and gather feedback on the UI/UX. The educational sandbox will

be finalised and populated with a lightweight deepfake generation algorithm (possibly using an open-source face swap library). By following our weekly schedule diligently, we anticipate having a fully functional, end-to-end Deepfake Detection and Education System ready for formal evaluation by March 2026. In summary, the project is on schedule and meeting its milestones: the foundational research and design are complete, development is well underway with early components already assessed locally, and we have a clear plan for the remaining implementation stages and deployment. We will continue to refine the system with iterative testing, all while adhering to the overarching goal of delivering a robust, transparent, and educational deepfake verification framework grounded in the latest research and real-world needs.